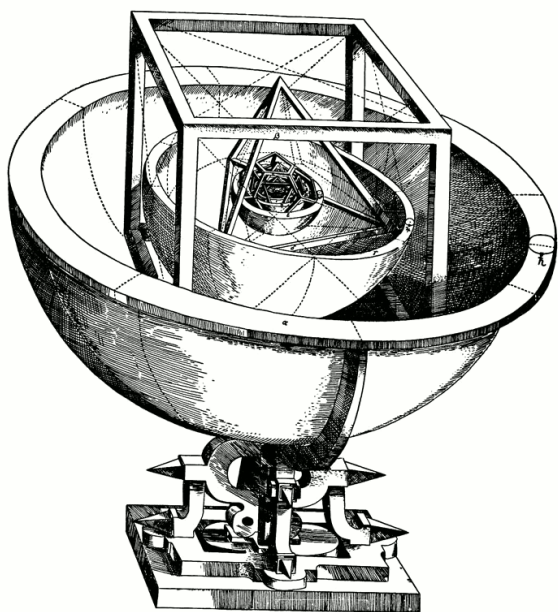


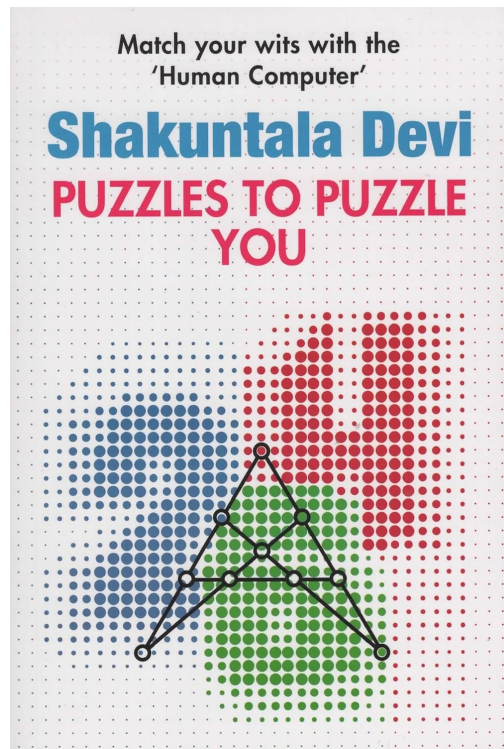
Puzzles to Puzzle You (Remastered)

Puzzles to Puzzle You (Remastered) is a monthly newsletter by Rani Rachavelpula, inspired by a popular book of the same name written by the ‘Human Computer’ Shakuntala Devi. Devi (1929-2013) was an Indian mental calculator, astrologer, and writer. She appeared in the 1982 *Guinness Book of World Records* for multiplying two 13-digit numbers in 28 seconds.¹ She also wrote *The World of Homosexuals*, which is considered the first study of homosexuality in India.

My mother gifted me her book for my eighth birthday. I took it everywhere and showed it to everyone. It didn’t take long for me to consider it the worst birthday present ever. My frustration was complex. I wasn’t agitated by failure. As a competitive and resilient child, I had experienced many ego deaths by age eight. Notably, my irritation grew most *after* learning the solutions. The reason: Devi—a beautiful and noble woman my home country adored—was a Mechanical Turk lacking real mathematical insight. Her puzzles were austere and unilluminating, inelegant and unintuitive. No human being should be a calculator.



Kepler’s Model of the Universe As Nested Platonic Solids (1596)



Devi’s Famed 1979 Book and the Inspiration for this Newsletter

This newsletter intends to do justice to the title of Devi’s book, not its content. My approach to puzzles is a Kantian one, inspired by his remarks on the mathematical sublime. I believe that a good puzzle is capable of producing a feeling of ‘supersensible comfort.’ It is the same feeling we experience at the foot of a mountain, engulfed in awe by its seemingly infinite size and our inexplicable ability to perceive the whole mountain *all at once* as a single mental picture. Both the faculty of imagination and the faculty of understanding must come together.

Some things I will look for when sourcing puzzles: visual beauty, multiple solutions, connecting the very concrete with the very abstract, symmetry, spontaneous synchronization, things that puzzled the ancients, carpentry, chirality, proofs by construction, gestalt, optical illusions, experimental mathematics, Donald Coxeter...

¹In 1980 at Imperial College London, Devi computed the following in 28 seconds: $7,686,369,774,870 \times 2,465,099,745,779 = 18,947,668,177,995,426,462,773,730$.

Technicalities

Form This newsletter adopts the format of the typical weekly crossword. Every month, I give a puzzle. The following month, a detailed solution is given along with that month's puzzle. I will also discuss any related mathematical concepts and/or whimsical applications.

Difficulty A motivated highschooler should be able to do the puzzles. Tenacity is necessary. Specialized mathematical knowledge is not. Many people these days loathe mathematics because they did not encounter it at the right level of difficulty. A good problem is non-trivial. It should require some thought, spanning anywhere between a single leisurely afternoon to a few weeks.² You may even need to do a little bit of research or review long-forgotten mathematical concepts. That being said, the puzzle should be doable. Even if you are not able to solve it in its entirety, some amount of progress should be possible. As Kant reminds us, it is delightful to feel the faculties of our mind expand and grow more powerful.³

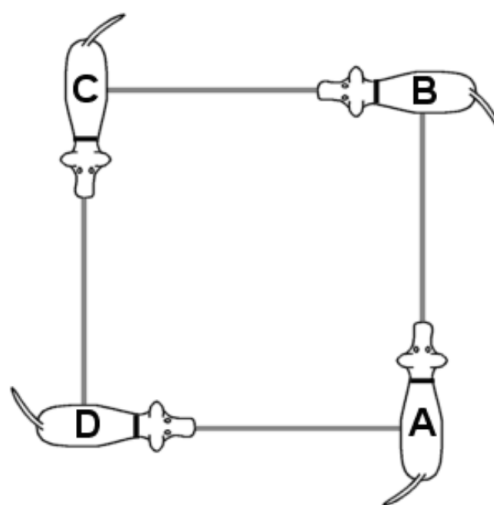
Intended Audience *This newsletter is for striving players, not achievement players.* An achievement player plays to win. A striving player temporarily acquires an interest in winning for the sake of the struggle. No one is giving you a grade. No one is checking your work. No one has to know you even attempted the puzzle. If you solve a puzzle, please do not spoil it for anyone else. Please grant everyone the equal opportunity to deliver a solution from their own exercise of reason.

Community Feel free to discuss with friends and/or submit your own puzzles. Mathematics is about drawing connections, of all possible kinds! If you come across a beautiful solution, do send it to me so I can include it in the following newsletter.⁴

Puzzle #1 (Dogs in Pursuit)

Dogs in Pursuit Four dogs are located at the corners of a square. On command they start moving with constant speed v , each towards its counterclockwise neighbor. If each side of the square has length l ,

1. How long will it take for the dogs to meet?
2. What distance will each have covered?



The puzzle can be generalized to arbitrary n -gons. It can further be generalized to any kind of animal capable of free motion and having an interest in its own kind so long as all the animal actors move at the same constant speed.



Four Turtles in Pursuit

²Of course, most of this time will be passive as you walk to work or twiddle your thumbs.

³Please keep in mind that I do not know the puzzle's solution when it is first sent out. I am not responsible for any errors in my judgments of difficulty.

⁴To avoid spoiling the puzzle for me, please warn me if your message contains a solution.